Smart Media Pricing (SMP): Non-Uniform Packet Pricing Game for Wireless Multimedia Communications

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***Abstract*—Service Providers (SPs) typically set data price according to flat rate, the time or the usage. However, the importance of multimedia content is different among various packets, and such diversity has largely been ignored in the past. In this paper, we propose a fundamentally new framework named Smart Media Pricing (SMP), to allocate price according to multimedia packet importance in terms of distortion reduction. Specifically, we investigate the profit maximization problem of a network SP (selling video transmission service) and a client device (requesting video sequences). The transmission power and the packet price are allocated non-uniformly among video frames. By deriving game equilibrium, the SP and the client device both can achieve maximum utilities. Simulation results show that the video quality, the energy efficiency, and the social profit are improved significantly compared to traditional uniform data pricing.**

***Index Terms*—Smart media pricing (SMP), wireless multimedia transmission, Stackelberg game.**

I. Introduction

The Cisco Visual Network Index predicted that global mobile data traffic will increase nearly tenfold between 2014 and 2019, and mobile video traffic will account for 72% of the total mobile data traffic by 2019 [1]. It will be enormous strain on already-overburdened mobile networks to meet this big multimedia data demand, which drives mobile network operators to innovate new pricing schemes. Smart data pricing (SDP) [2] is likely to emerge as an effective way to influence the user behavior in utilizing network resources. However, multimedia smart pricing has largely been ignored in literature. In this paper, we propose a fundamentally new framework named Smart Media Pricing (SMP) to allocate different prices to different multimedia packets with various quality contributions.

Traditionally, network operators have used simple flat-rate broadband data plans for both wired and wireless network access. Flat-rate fee results in traffic congestion at peak hours. With the popularity of bandwidth-hungry mobile devices, and exponential growth of data-hungry apps, mobile videos, and cloud-based game services, wireless Service Providers (SPs) are gradually moving toward more sophisticated pricing schemes [3]. Over the past few years, SPs around the world have started to offer innovative pricing plans, including usage-based and app-based pricing to tackle the problem of network congestion [4]. SDP plans that have been proposed in the past or are being explored include: (a) Usage-based pricing [5], (b) App-based pricing and sponsored access [6], (c) Quota-aware content distribution [7], and (d) Day-ahead dynamic pricing [8]. Though static pricing is the dominant pricing strategy in today's market, the prices should be dynamically updated to cope with the high-quality multimedia data demand.

Pricing plans like time dependent pricing (TDP) were proposed to help reduce the network's peak-to-average usage ratio while benefiting both operators and customers, by encouraging users to shift a part of their demands to off-peak time [8], [9]. However, such mechanisms typically require user preferences about shifting demands, which may be inaccurate or difficult to obtain due to privacy concerns. To address this issue, a fixed-budget rebate mechanism was proposed in [10] that gives each user rewards proportional to the percentage of contribution to the aggregate reduction in peak-time demand. Ren and van der Schaar [11] explored the interaction of dynamic pricing and service request scheduling, and demonstrated that jointly optimizing these two decisions could significantly enhance the wireless cloud SP's long-term profitability. Kun and Hossain [12] proposed a dynamic Stackelberg game framework to jointly address the problems of spectrum partitioning and user-controlled mode selection. Andrews *et al.* [6] built a proper contractual relationship between the SP and content provider through a Stackelberg game. A smart mobile data pricing testbed with speed tiers was proposed in [13]. However, most of the study above considered general data networks and overlooked rate-distortion characteristic of a video stream, which can be exploited to maximize the video quality rather than the bit error rate or throughput.

Current simple policies like usage-based pricing (byte-counting) force users to pay the same amount per unit of bandwidth consumed irrespective of the service quality, and fail to account for the fact that different frames with unequal

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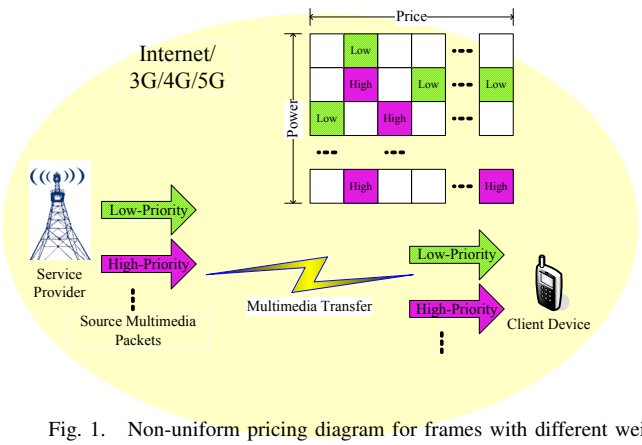

Fig. 1. Non-uniform pricing diagram for frames with different weights.

importance. Different multimedia packets have different bandwidth requirements and communication energy consumption to attain a certain Quality of Service (QoS) for the users. A simple differentiated pricing plan is Paris metro pricing, in which the SP separates data traffic into different logical traffic classes and charges different prices for logically separate traffic class. Only users willing to pay a higher price will adopt a class leading to better QoS. Although it allocated prices according to the fixed traffic classes, the fundamental diversity among frames for a single video sequence was not considered.

In our previous work [14] we have developed a game model to optimize power purchasing strategies from multimedia sources in device-to-device communications. The game model was established between several source nodes and the BS, but pricing was only considered uniformly for different packets. Different from our previous work, in this paper we propose a new SMP framework focusing on non-uniform pricing for different multimedia packets between one client and one SP. This paper aims to balance three interactive components: 1) the operator's cost of delivering contents to the client device, 2) the service QoS, and 3) the amount that the user is willing to pay. We assume that the SP has access to the content source such as YouTube, Netflix, CNN, Facebook, etc. It also controls over the power allocation of base station in networks like AT&T and Verizon. The SP has interest in conflict with that of customers. It seeks to charge more while the clients seek to pay less. The transmission power is taken into account as an operating cost to ensure a surplus budget in supporting this resource-consuming application.

II. System Model of SMP

Given the flexibility to change the selling price, the fundamental question is: *What is the optimal dynamic media pricing policy for a SP, in terms of maximizing the expected profit?* Such network interaction can be analyzed as a Stackelberg game, where the SP is the leader and the client user is the follower. A server naturally wishes to set a higher price to get a higher revenue margin; yet in doing so, it also bears the risk of discouraging the customer's resource demand.

We consider a wireless multimedia transmission network presented in Fig. 1, in which a client device requests a video sequence from a SP. We follow the microeconomic theory concept introduced in [3]. Consider a last-mile SP that wishes to build a single access link serving a customer. The SP will charge each frame (or packet, we will use frame and packet interchangeably through this paper) based on the amount of transmission power reserved by this user on the shared access link. And the price is also allocated according to the multimedia characteristics of the frame, such as distortion reduction and bit length. Hence, the SP aims to determine: (1) the optimal transmission power P_i for each packet i, and (2) the quality-based price λ_i for each packet i. The game between a client and a SP is described as follows.

A. Optimization on the Client (Buyer) Side

The client, as a follower of the game, is the buyer of requiring the video stream including N frames. For each frame with different importance, the client performs the Stackelberg game with the SP. The client aims to achieve the best performance using the least reimbursement cost. Let D_i and L_i denote the distortion reduction and the bit length of the ith frame. The set of ancestor frames, which the ith frame refers to, is denoted by $\mathcal{A}_i$. For each frame, its quality is related to its distortion reduction, the successful transmission probability of itself, and the successful transmission probabilities of its ancestor frames:

$$Q_i = D_i(1-e_i)^{L_i} \prod\nolimits_{k\in\mathcal{A}_i} (1-e_k)^{L_k} \tag{1}$$

where e_i is the bit error rate (BER) of frame i.

We use logarithmic function to define the benefit utility. Let λ_i and P_i denote the price per unit of power and the transmission power sold from the SP. The product of the price per unit of power and the transmission power is the payment from the client. The benefit utility of this client can be expressed as:

$$U_i = a\log_2 Q_i - \lambda_i P_i \tag{2}$$

where a denotes the currency gain per logarithmic scale of video quality. The optimization problem for the client is to set proper amount of buying power that maximizes its utility, subject to the power constraint:

$$\begin{aligned} \max U_i &= a\log_2\left(D_i(1-e_i)^{L_i}\prod\nolimits_{k\in\mathcal{A}_i}(1-e_k)^{L_k}\right) - \lambda_i P_i \\ \text{s.t. } & P_{\min} \le P_i \le P_{\max} \\ & U_i \ge 0. \end{aligned} \tag{3}$$

The constraints $P_{\min}$ and $P_{\max}$ are the minimal power that guarantees the transmission quality and the maximal transmission power allowed by the SP. The utility of a user should also be non-negative when finding the best power and price responses. The BER e_i in equation (3) has a close-form mathematical relationship with the transmission power

when we consider Quadrature Phase Shift Keying (QPSK) modulation scheme [15]

$$e_i = \frac{1}{2}\text{erfc}\left(\sqrt{\frac{P_i g}{2}}\right) \tag{4}$$

where $P_i g$ is the signal-to-noise ratio (SNR) with $g = G/(R_s N_0)$. G is the channel quality between the SP and the client user, R_s is the symbol rate, and N_0 is the noise power density. Higher density modulation schemes such as Quadratic Amplitude Modulation (QAM) can be seamlessly applied to this model.

B. Optimization on the SP (Seller) Side

Being a rational player, the SP knows how much power the follower will buy with given prices. Consequently, the SP can set its optimal prices according to the power responses of the client device. The benefit of the SP is the money that the client user pays. The utility function of the SP can be expressed as

$$U_{\text{SP}} = \lambda_i P_i - c\left(P_i, L_i\right) \tag{5}$$

where $c\left(P_i, L_i\right)$ is the cost function. A simple definition of it is the energy cost determined by the transmission power P_i and the packet length L_i, i.e.,

$$c\left(P_i, L_i\right) = \frac{\mu P_i L_i}{R_i} \tag{6}$$

where μ and R_i denote the transmission cost per unit of energy consumption and the transmission rate, respectively. The optimization problem of the SP is to set a proper charging price that maximizes its utility, subject to the cost constraint:

$$\begin{aligned} \max U_{\text{SP}} &= \lambda_i P_i - \frac{\mu P_i L_i}{R_i} \\ \text{s.t. } \lambda_i &> \frac{\mu P_i L_i}{R_i} \\ U_{\text{SP}} &\geq 0. \end{aligned} \tag{7}$$

III. Analysis of Non-Uniform Multimedia Pricing Game

A. Stackelberg Equilibrium

Equilibrium is a stable state of a game where the SP and the client device both achieve the optimal utility and will not deviate as long as they are rational players. To derive the Stackelberg equilibrium, we need to solve the utility optimizations of the SP and the client device.

Property 1: The utility function of the client U_i when downloading the ith packet is concave in power that the client device wants to buy, when λ_i is fixed and $P_{\min} \leq P_i \leq P_{\max}$.

Proof: The utility function U_i is continuous versus P_i when the price λ_i is fixed. Let $\alpha = a * g/(4\sqrt{\pi}\ln 2)$. Taking the first derivative of the utility function U_i in (3), we have

$$\frac{\partial U_i}{\partial P_i} = \frac{\alpha L_i \exp\left(-P_i g/2\right)}{\left(1-e_i\right)\sqrt{P_i g/2}} - \lambda_i, \tag{8}$$

and the second order derivative is

$$\begin{aligned} \frac{\partial^2 U_i}{\partial P_i^2} =& \frac{\alpha g L_i \exp\left(-P_i g/2\right)}{2\left(1-e_i\right)\sqrt{P_i g/2}} \times \\ & \left(-1 - \frac{\exp\left(-P_i g/2\right)}{2\sqrt{\pi}\left(1-e_i\right)\sqrt{P_i g/2}} - \frac{1}{P_i g}\right). \end{aligned} \tag{9}$$

Since α, g, $\exp(\cdot)$, and $(1-e_i) > 0$, then we have $\partial^2 U_i/\partial P_i^2 < 0$. Therefore, the utility function for transmitting the ith frame U_i is concave with respect to P_i when the price λ_i is fixed. ■

Due to *Property 1*, the optimal transmission power can be acquired by letting the first order derivative of utility function U_i in (8) equal to zero. Then we have

$$\frac{\alpha L_i \exp\left(-P_i g/2\right)}{\left(1-e_i\right)\sqrt{P_i g/2}} - \lambda_i = 0. \tag{10}$$

Equation (8) may have more than one root, we only choose the one obtaining the largest utility U_i with the lowest transmission power. It indicates that the user's power response has a fixed relationship with given price, which can be expressed as $P_i^*\left(\lambda_i\right)$. For each given price, the best power response can be obtained using Newton methods by tools like MATLAB. Note that the power is constrained as $P_{\min} \leq P_i \leq P_{\max}$, and then the global optimum that maximizes the client's utility function U_i can be searched in the set $\{P_{\min}, P_i^*\left(\lambda_i\right), P_{\max}\}$. Thus, we have the maximum and minimum prices expressed as

$$\begin{aligned} \lambda_{\min} &= \frac{\alpha L_i \exp\left(-P_{\max} g/2\right)}{\left(1-e_i\right)\sqrt{P_{\max} g/2}} \\ \lambda_{\max} &= \frac{\alpha L_i \exp\left(-P_{\min} g/2\right)}{\left(1-e_i\right)\sqrt{P_{\min} g/2}}. \end{aligned} \tag{11}$$

We denote the optimal response of the client user as

$$P_i^{\text{opt}}\left(\lambda_i\right) = \begin{cases} P_{\max}, \lambda_i \leq \lambda_{\min} \\ P_i^*\left(\lambda_i\right), \lambda_{\min} < \lambda_i < \lambda_{\max} \\ P_{\min}, \lambda_i \geq \lambda_{\max} \end{cases}. \tag{12}$$

With the fact that the SP knows the optimal power responses $P_i^{\text{opt}}\left(\lambda_i\right)$ of the client device, the utility function of the SP in (7) can be rewritten as

$$U_{\text{SP}}\left(\lambda_i\right) = \left(\lambda_i - \frac{\mu P_i L_i}{R_i}\right) P_i^{\text{opt}}\left(\lambda_i\right). \tag{13}$$

Lemma 1: A real function which is differentiable must be a continuous function. [16]

Lemma 2: A continuous real function on a closed interval must contain a maximum value and a minimum value. [16]

Property 2: The utility function of SP U_{SP} has a maximum value on the closed interval $[\lambda_{\min}, \lambda_{\max}]$.

Proof: Taking the first derivative with respect to λ_i in (10), we have

$$\frac{\partial P_i}{\partial \lambda_i} = \frac{-2\left(1-e_i\right)\sqrt{P_i g/2}}{\alpha g L_i \exp\left(-P_i g/2\right)\left(1 + 1/(P_i g) + \lambda_i/(2\sqrt{\pi}\alpha L_i)\right)} \tag{14}$$

Taking the first derivative of the utility function of SP in (13), we have

$$\frac{\partial U_{\text{SP}}}{\partial \lambda_i} = P_i + \left(\lambda_i - \frac{\mu P_i L_i}{R_i}\right)\frac{\partial P_i}{\partial \lambda_i}. \tag{15}$$

Combining (14) and (15), the function U_{SP} is proved to be differentiable. Because of *Lemma 1*, we have that the function U_{SP} is continuous. Together with *Lemma 2*, we prove that the utility function of SP U_{SP} has a maximum value on the closed interval $[\lambda_{\min}, \lambda_{\max}]$.

■

Thus, we can find the optimal price response of the SP by maximizing the utility function U_{SP}. The optimization problem can be solved using genetic algorithm or global searching. We denote the optimal price response as $\lambda_i^* = \arg\max\{U_{\text{SP}}\}$. The global optimal price for the ith frame can be search in the set of $\{\lambda_{\min}, \lambda_i^*, \lambda_{\max}\}$. It is the Stackelberg price equilibrium λ_i^{SE}, with $\lambda_i^{\text{SE}} \in \{\lambda_{\min}, \lambda_i^*, \lambda_{\max}\}$. Overall, we have the following theorem.

Theorem 1: The Stackelberg power equilibrium P_i^{SE} and Stackelberg price equilibrium λ_i^{SE} can be expressed as

$$\lambda_i^{\text{SE}} \in \{\lambda_{\min}, \lambda_i^*, \lambda_{\max}\} \tag{16}$$

$$P_i^{\text{SE}} = \begin{cases} P_{\max}, \lambda_i = \lambda_{\min} \\ P_i^*(\lambda_i^*), \lambda_i = \lambda_i^* \\ P_{\min}, \lambda_i = \lambda_{\max} \end{cases}. \tag{17}$$

The SP and the client user will not deviate from the equilibrium power and equilibrium price, wherein they achieve maximum utility. Summing up the utility of the SP and the client device, we have the social profit of the whole network. The social profit is also maximized when the game reach equilibrium.

B. Non-Uniform Media Pricing Algorithm

The proposed non-uniform pricing scheme is described in Algorithm 1. For each single packet, the Stackelberg game interactions are performed once. The optimal power and price can be decided based on *Theorem 1*. For each possible price the SP gives, the client device can obtain the best power response using Newton's Method, where $[J_F(P_n)]^{-1}$ is the left inverse of the Jacobian matric $J_F(P_n)$ of F evaluated for P_n. Then the SP can achieve maximum utility by setting appropriate price. They are the Stackelberg equilibria wherein the SP and the client device both get their maximal utilities.

The computing complexity for the proposed SMP scheme is $O(NM)$, which is decided by the account of packets N and the maximum iteration steps M to find the optimal value. Since the equilibrium power and the equilibrium price are variable with the distortion reduction and the bit length of a packet, we can make a two-dimension table off-line in advance. The searching table can be updated with the dynamic estimated channel quality in spare time periods between transmitting video sequences. Then, the optimal power and price can be searched directly from the table when non-uniform SMP was performed. The computing complexity during communication can be reduced a lot.

Algorithm 1 : Non-Uniform Media Pricing

Initialization:

Initialize the currency gain a, unit cost c, channel quality G, frequency bandwidth W, symbol rate R_s, noise power density N_0, maximum iteration steps M, and distortion reduction D_i and length $L_i, i \in [1, N]$.

Iteration:

```
while i++ ∈ [1, N] do
    Set D = D_i and L = L_i;
    Initialize U_max = 0;
    Let χ = linspace(λ_min, λ_max, M);
    while m = 1 : M do
        Set λ = χ(m) and initialize P_0 = 0;
        Let F(P) = αL exp(−Pg/2) / ((1 − e)√(Pg/2)) − λ;
        while |P_{n+1} − P_n| > 10^{−10} do
            Set P_{n+1} = P_n − [J_F(P_n)]^{−1} F(P_n);
            n = n + 1;
        end while
        Set P* = P_{n+1};
        Calculate the utility of the SP U_SP;
        if U_SP > U_max and U_i ≥ 0 then
            Set U_max = U_SP;
            Set λ_i^SE = current λ and P_i^SE = P*;
        end if
    end while
end while
```

Output:

The optimal power P_i^{SE} and the optimal price λ_i^{SE} for each packet $i \in [1, N]$.

IV. Simulation Results

A. Simulation Parameters

The video sequence for simulation is "*Foreman*" with H.264 coder. The distortion reduction and the reference relationship are calculated by JM 18.0 Reference Software. The intra frame frequency of the video is 3, i.e., the frame sequence is "IPPIPPIPP..." Other intra coding rates can also be applied to this algorithm. The I-frame's successful transmission only relies on itself, while the P-frames refer to the previous I-frame and P-frames. We set the frequency bandwidth as 15 MHz and the maximum iteration steps as 200. The channel quality attenuation between the SP and the client user is set to be $G = -75$ dB, the symbol rate is $R_s = 1$ kHz, and the initial noise is $N_0 = -106$ dB/Hz. The maximal and minimal transmission power for the SP is $P_{\max} = 48$ dBm and $P_{\min} = 25$ dBm. The initial values for a and μ are 5 and 1.

The proposed Non-Uniform Pricing scheme (Non-UP) smartly allocates different prices and transmission power to frames with various multimedia characteristics, i.e., the distortion reduction and the bit length. We also utilize two comparison schemes, which are Uniform Pricing based on Equal frame Importance (UPEI) and Uniform Pricing based on Equal transmission Power (UPEP), respectively. In UPEI, each frame uses the same distortion reduction and bit length, and no importance difference is considered. In UPEP, each frame is delivered with equal power, and their power may change with the various video characteristics.

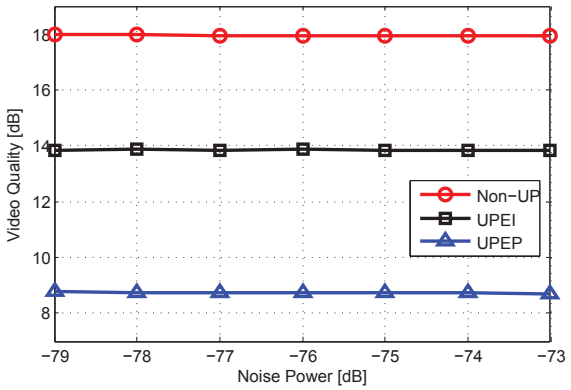

Fig. 2. Comparison of video quality under various noise power.

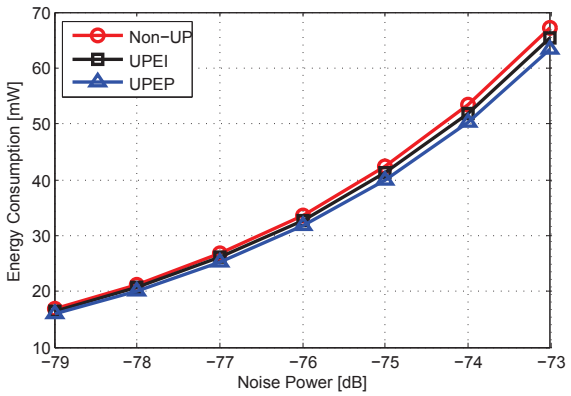

Fig. 3. Comparison of energy consumption under various noise power.

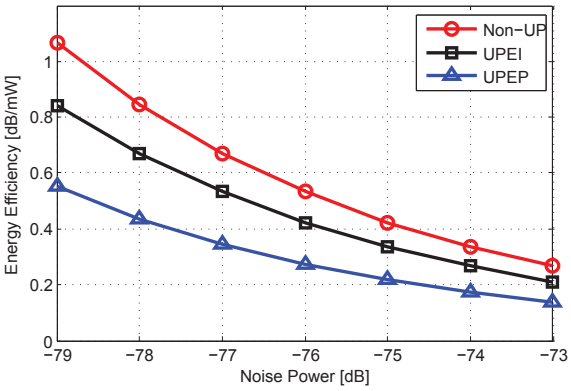

Fig. 4. Comparison of energy efficiency under various noise power.

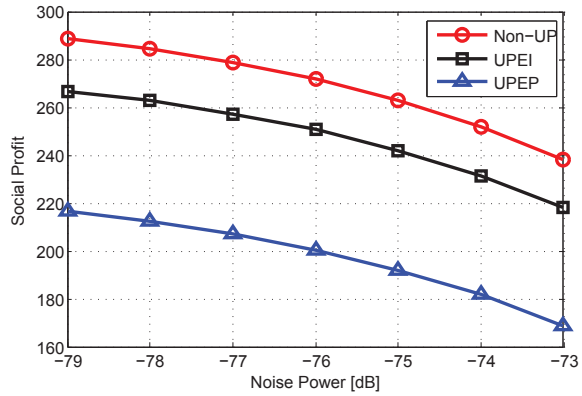

Fig. 5. Comparison of social profit under various noise power.

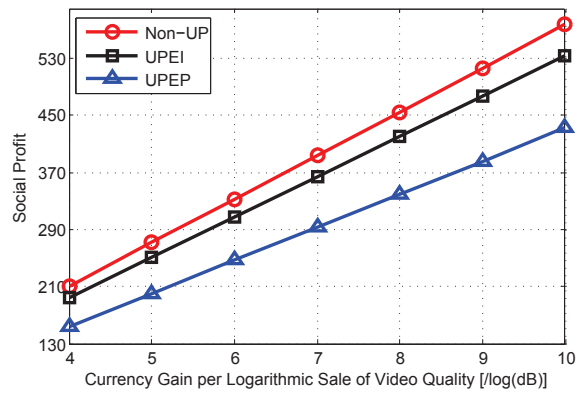

Fig. 6. Comparison of social profit with various currency gain per logarithmic sale of video quality a.

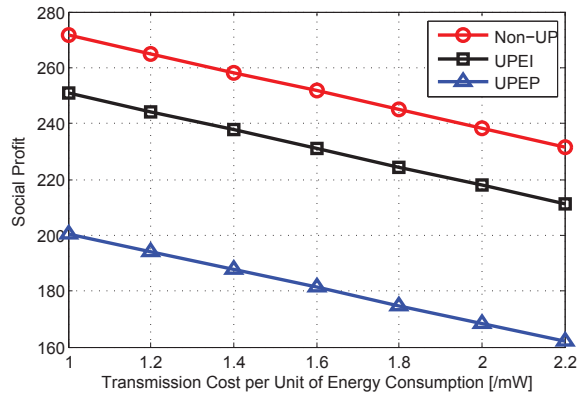

Fig. 7. Comparison of social profit with various transmission cost per unit of energy consumption μ.

B. Performance

Figs. 2-5 demonstrate the comparison of video quality, energy consumption, energy efficiency, and social profit of these three schemes. The energy efficiency is defined as the video quality per unit of energy consumption. Given various noise power, the proposed non-uniform pricing scheme outperforms the uniform pricing schemes significantly. The video quality is significantly enhanced with almost the same power consumption. It is because those more important video frames are transmitted with higher power and at higher price, while the unimportant frames are delivered at lower price with lower power. The social profit is increased at the same time by allocating unequal prices among frames using the proposed non-uniform SMP scheme.

Figs. 6-7 show the social profit gain when we change the currency gain per logarithmic scale of video quality a and the transmission cost per unit of energy consumption μ. Social profit is the sum of utilities of the SP and the client device. All the social profits of the three schemes increase with larger a or smaller μ linearly. Thus, the different initializations of a and μ do not affect the conclusion that the proposed non-uniform scheme always achieves the highest social profit.

V. Conclusion

In this paper, we introduced a new framework named SMP and proposed a new non-uniform media pricing scheme using Stackelberg game. The game was developed between a SP which sold video services and a client device which requested a video sequence. The utilities of the SP and the client device were both optimized when we achieved the game equilibrium. The transmission power and the selling price were also set unequally among video frames with various media characteristics. Simulation results showed that our proposed non-uniform pricing scheme improved the performance significantly compared to traditional uniform pricing schemes. The proposed multimedia pricing framework can be extended to analyze SMP between several client users and a SP, or between several client users and several SPs.

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